



Document Management System

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Effective Date: 16/12/2024 Target Review Date: 31/12/2027											
Title: Sepsis & Septic Shock Management											
Early Recognition of Sepsis in Children What is sepsis? Sepsis remains the primary cause of death from infection despite advances in modern medicine, including vaccines, antibiotics, and acute care with hospital mortality rates between 30 and 60%. In addition it is recognized as a leading cause of morbidity and mortality in children worldwide. It is defined by international consensus guidelines as the presence of systemic inflammatory response syndrome plus suspected or proven infection. Sepsis is a clinical syndrome, which results from dys-regulation of systemic inflammatory response to infection. This is characterized by a generalized pro-inflammatory cascade, which may lead to widespread tissue injury. A. Progression of clinical features Infection → Systemic inflammatory response (SIRS) → Sepsis → Severe Sepsis → Death. <table border="1"> <thead> <tr> <th>Infection</th> <th>SIRS</th> <th>Sepsis</th> <th>Severe Sepsis</th> <th>Death</th> </tr> </thead> <tbody> <tr> <td colspan="5"> </td> </tr> </tbody> </table> Presence of two or more of the following symptoms (See Table 1):		Infection	SIRS	Sepsis	Severe Sepsis	Death					
Infection	SIRS	Sepsis	Severe Sepsis	Death							

- Temperature instability core temp > 38.5 / < 36
- Tachycardia or bradycardia
- Tachypnea
- White blood cell count $> 12,000/\text{mm}^3$ or $< 4,000/\text{mm}^3$ or $> 10\%$ immature neutrophils

SIRS can be caused by infection, trauma, burns, pancreatitis, and other insults which result in tissue damage

Table 1. Age Specific Vital and WBC

Age Group	Tachycardia	Bradycardia	Respiratory rate	WBC $\times 10^3/\text{mm}^3$	Systolic BP mm Hg
0 days – 1 wk	> 180	< 100	> 50	> 34	< 65
1 wk – 1 mo	> 180	< 100	> 40	> 19.5 or < 5	< 75
1 mo – 1 yr	> 180	< 90	> 34	> 17.5 or < 5	< 100
2-5 yrs	> 140	NA	> 22	> 15.5 or < 6	< 94
6-12 yrs	> 130	NA	> 18	> 13.5 or < 4.5	< 105
13 - < 18 yrs	> 110	NA	> 14	> 11 or < 4.5	< 117

Infection SIRS Sepsis Severe Sepsis Death

SIRS with a presumed or confirmed infectious process

Sepsis with signs of at least one acute organ dysfunction

- Respiratory
- Cardiovascular
- Renal
- Hepatic
- Haematological
- Change in mental status
- Unexplained metabolic acidosis

Septic shock

Severe sepsis with hypotension refractory to adequate volume resuscitation

RED FLAGS indicating severe sepsis (at presentation):

1. Rapid deterioration in less than 6 hours
2. Poor or no response to treatment
3. WCC less than $2 \times 10^9/L$ and/or neutropenic or WCC more than $30 \times 10^9/L$
4. Platelets low (less than 50)
5. Lactate $> 2 \text{ mmol/L}$ (arterial or venous or capillary)
6. Hypoglycemia
7. Hypothermia
8. Altered or variable conscious level (check with parents/carriers)
 - Quiet submissive child (not usual self/behavior)
 - Aggressive, combative child (not usual self/behavior)
9. Oliguria or anuria for 6-12 hours
10. Deranged liver function
11. Central venous saturation less than 70% (or 20-25% less than usual SpO_2 in cyanotic heart disease)

1. Male sex: severe sepsis is 15% more common in boys with an increased incidence in boys under 4 years
2. Underlying disease/Co-morbidities
 - Neonates: Gram negative sepsis especially in those with probable Galactosaemia; Group B Strep; HSV
 - Infants with cardiovascular or respiratory disease
 - Older Child with Neuromuscular disease or Oncology diagnosis
 - Any age: Immunodeficiency
3. Recent Illness
 - Concurrent/recent Chicken Pox (within last 6-8 weeks) – Invasive Group A Strep & *Staph. aureus*
 - Post Influenza A or B – *Streptococcus pneumoniae*, *N. meningitidis*, Invasive Group A *Streptococcus*, *Staphylococcus aureus*
4. Chronic Illness patients (NB may be carriers of: *Pseudomonas* or MRSA or other multi-resistant organisms)

B. Management

Recognition:

Flash capillary refill and altered mental status with a febrile illness are the hallmarks. Rash in this setting should signal possible meningococemia. Increased suspicion is warranted in children with co-morbidities.

The task force has identified **two new clinical criteria** that clinicians should use in diagnosing patients with septic shock. These include:

- ✓ Persisting hypotension requiring vasopressors to maintain MAP appropriate for age (Table 1.)
- ✓ Blood lactate >2 mmol/L despite adequate volume resuscitation

Resuscitation:

The ABCDs of sepsis:

A – Airway

B – Breathing

C – Crystalloids,

D – Dextrose, Drugs = antibiotics, inotropes

S – Steroids

Airway:

- ✓ Oral endotracheal tube – Consider cuffed ETT if any evidence of pulmonary edema.

- ✓ Placenasogastric tube to decompress the stomach.
- ✓ Chest X-ray to assess endotracheal and nasogastric tubes positions.
- ✓ End-tidal CO₂ monitor to be connected within circuit if available

Triggers for Intubation / Anaesthetic Review

1. Decreased conscious level (i.e. $GCS \leq 8$, $AVPU \leq P$) OR fluctuating conscious level
2. Increasing respiratory failure, signs of exhaustion
3. Impending cardiovascular collapse
4. Fluid refractory shock
 - 40-60 mL/kg resuscitation fluid given within the 1st hour without reversal of shock
 - Increasing size of liver
 - Requirement for multiple inotrope infusions

Breathing:

- ✓ Place on ventilator with age appropriate settings aiming for tidal volume 5-8 ml/kg.
- ✓ Maintain positive end expiratory pressure (PEEP);
- ✓ High PEEP in pulmonary edema up to 10.
- ✓ Monitor end tidal CO₂ (Permissive hypercapnia accepting pH 7.27)
- ✓ If FiO₂ >60% accept SpO₂ up to 89%-90% and once FiO₂ < 60% accept saturation >94%.
- ✓ Do not expect 100% saturation in a sick child.

Circulation:

- ✓ At least one good peripheral access or Intraosseous line (can be used as central access)
- ✓ Insert and secure multiple lumen central venous catheter (CVC) if possible and confirm position by CXR for neck line and abdominal X-ray for femoral line.
- ✓ Measure central venous pressure and saturation.

✓ If needed start inotropes through peripheral line or Intraosseous eventually aiming for CVC (may not be achievable until received adequate fluid resuscitation).

✓ Insert urinary catheter and monitor urine output.

Drugs:

✓ Assure antibiotic is given within first hour of resuscitation.

✓ Steroids:

- Catecholamine resistant shock,

- If patient has signs of meningitis,

- or if patient has been on steroids

- Consider if patient on steroids within last year or at high risk for being steroid deficient (eg; oncology, organ transplant, Asthma, meningococemia (Waterhouse-Friderichsen syndrome), congenital adrenal hyperplasia)

✓ Start maintenance intravenous fluid containing dextrose to maintain blood glucose level

✓ Antibiotics:

Antibiotics to be given within first 1 hr

➤ Neonates/infants < 2 months

- Ampicillin 100mg/kg/day every 6 hr IV/IM + gentamicin 7.5mg/kg/day every 8hr IV/IM

- Or Cefotaxime 50mg/kg/dose every 6 hr IV

➤ Infants and children > 2 months

- **(Immune-competent):** Ceftriaxone + 50mg/kg/dose every 12 hr IV

- **(Immunosuppressed):**

- Cefepime 50mg/kg every 12 hr + gentamicin

- Or Tazocin 200-300mg/kg/day every 6-8hr IV + gentamicin

➤ **If Encephalopathic:** Add acyclovir

Investigations:

- ✓ Blood cultures X 2
- ✓ Sputum culture
- ✓ Urine microscopy, analysis and culture
- ✓ Complete blood count (CBC), renal function test, liver function test, coagulation and C-reactive protein
- ✓ Blood gas (arterial, venous or capillary)
- ✓ Serum lactate level
- ✓ Blood glucose level
- ✓ If hemodynamically stable send CSF culture

Communication:

- ✓ Parents: Outline diagnosis, management and prognosis.
- ✓ Maintain contact with senior for update and ongoing advice.
- ✓ If case is referred to tertiary hospital, follow up advice and update the referred centre.
- ✓ Document all communicated information in the system for continuation of the treatment to avoid conflicts.

Goal Directed Therapy

1. Maintain or restore airway, oxygenation & adequate CO₂ clearance.
2. Restore and maintain normal perfusion.
 - a. No difference in quality between central & peripheral pulses.
 - b. Warm extremities (caution may have warm shock).
 - c. Heart rate, blood pressure within normal limits for age.
 - d. Central capillary refill time < 2 seconds.
3. Normal mental status.
4. Urine output returned (>0.5 – 1 ml/kg/hour).
5. Serum lactate < 2 mmol/L.
6. Central venous saturation (S_{CV}O₂) > 70% from neck line (ideally).
7. Normal blood glucose concentration.

Criteria for Organ Dysfunction:

The new diagnostic tool is named quick SOFA, or qSOFA. It consists of three simple tests that clinicians can conduct at the bedside to identify patients at risk for sepsis. The qSOFA assessment directs physicians to look for these warning signs in patients:

- Alteration in mental status
- Decrease in systolic blood pressure of less than 100 mm Hg
- Respiration rate greater than 22 breaths/min



A. Bundles

To Be Completed Within 3 Hours Of Time Of Presentation*:

1. Measure lactate level
2. Obtain blood cultures prior to administration of antibiotics
3. Administer broad spectrum antibiotics
4. Administer 30ml/kg crystalloid for hypotension or lactate ≥ 3 mmol/L *

through chart review.

To Be Completed Within 6 Hours Of Time Of Presentation:

5. Apply vasopressors (for hypotension that does not respond to initial fluid resuscitation) to maintain a mean arterial pressure (MAP) age appropriate.
6. In the event of persistent hypotension after initial fluid administration or if initial lactate was ≥ 3 mmol/L, re-assess volume status and tissue perfusion
7. Re-measure lactate if initial lactate elevated.

If central line is in situ:

- a. Measure central venous pressure (CVP)
- b. Measure central venous oxygen saturation (ScvO₂)

Reference

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[Back](#)


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